Exploratory Data Analysis & Statistical Review

Executive Summary Report: NYC TLC Data Project

Prepared by Automatidata for New York City Taxi & Limousine Commission

Overview

The overall purpose of this project is to help the New York City Taxi and Limousine Commission (NYC TLC) better understand fare patterns and build predictive models to maximize driver revenue. This integrated report consolidates the initial Exploratory Data Analysis (EDA), the identification of anomalies in trip data, and the A/B statistical test on the relationship between payment method (credit card vs. cash) and total fare amount.

Problem

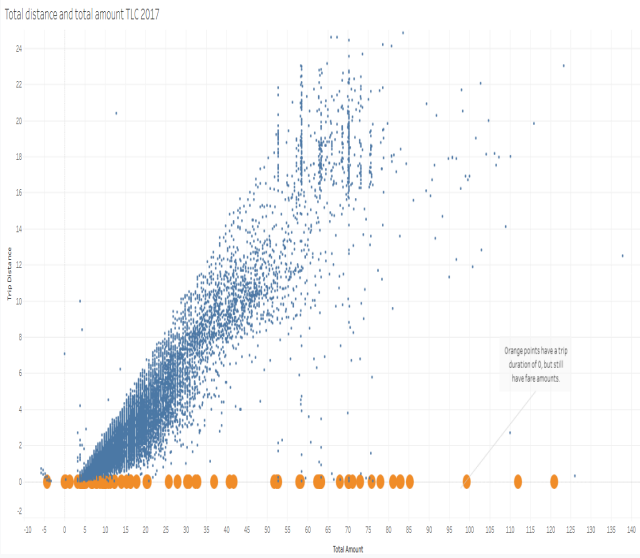
During the initial exploratory data analysis and data preparation phase, the Automatidata team identified two critical challenges:

- **Data Anomalies and Inconsistencies (EDA):** A significant volume of trip records was detected with a recorded distance of 0.0 miles, but with positive total fare charges. These outliers represent an obstacle to training accurate regression models.
- **Revenue and Payment Method Variability:** Drivers experience high fluctuation in tips and revenue depending on the payment method used by riders, requiring statistical evidence on whether promoting digital payments increases the amount collected.

Key Insights (EDA)

Exploratory Data Analysis Findings:

- **Distance vs. Amount Relationship:** The scatter plot shows an expected direct linear correlation between **trip_distance** and **total_amount**, but it reveals anomalous clustering at fixed amounts (e.g., flat rates to JFK Airport).
- **Outlier Identification:** Trips with total costs of up to $80+ were identified despite having a recorded distance of 0.0 miles. It is recommended to remove these from the regression training set.



See visualization here.

- **Payment Distribution:** Credit cards are the predominant payment method for long, high-value trips, whereas cash predominates for short trips.

Fare Statistical Summary:

Method	Average	Std. Dev
Credit card	$16.35	$12.50
Cash	$13.54	$10.80
Difference	**+$2.81**	**p < 0.001**

Details & A/B Test

Prior to statistical testing, exploratory data analysis revealed a heavy right-skewness in trip distances and fare amounts, alongside anomalous records (such as 0.0-mile trips with positive fares). Cleaning these outliers and understanding the underlying payment distributions ensured a reliable foundation for our hypothesis testing.

To determine whether the payment method statistically impacts the total amount charged, Automatidata performed a two-sample hypothesis test (Welch's t-test):

- **Null Hypothesis (H_0):** There is no difference in the average total fare between customers who pay with credit card and those who pay in cash ($\mu_1 = \mu_2$).
- **Alternative Hypothesis (H_a):** The average total fare for credit card customers is significantly different/higher than that of cash customers ($\mu_1 \neq \mu_2$).
- **Statistical Result:** The null hypothesis was rejected with an extremely low p-value ($p < 0.05$). Fares paid with credit card ($16.35) are statistically higher than those paid with cash ($13.54).

Next Steps

- **Business Strategy:** Encourage credit card payments by installing visible signage in cabs ('Credit card payments preferred') and instructing drivers to inform passengers of contactless card reader availability.
- **Data Cleaning for Modeling:** Filter out all records with a distance of 0.0 miles and a positive fare amount from the training dataset to prevent bias in the final predictive model.
- **Advanced Multivariable Analysis:** Implement multiple linear regression models or Propensity Score Matching to isolate the true effect of payment method while controlling for trip distance, time of day, and drop-off location.